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Tumor Location of Uveal Melanoma and Impact on Metastasis-Free Survival in 1001 Cases

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ABSTRACT

Background: There is limited literature on tumor location of uveal melanoma (UM) and the relative impact on metastasis-free survival (MFS).

Methods: A review of 1001 consecutive eyes with UM was performed and precise tumor location was identified according to quadrant (5 zones), clock hour (13 zones), anteroposterior (5 zones), and a combination of clock hour and anteroposterior regions (49 zones). Heat maps were constructed for frequency in each zone and Kaplan-Meier estimates for 5- and 10-year MFS was calculated.

Results: The UM quadrant regions included central (macula and juxtapapillary) (18%), superior (19%), nasal (18%), inferior (18%), and temporal (27%). The central region demonstrated smaller UM thickness ($p < .01$) and diameter ($p < .01$). The MFS (5- and 10-years) for central region (90%, 86%) was more favorable than other quadrants ($p < .01$).

The UM clock hour regions included central (18%), 1:00 (6%), 2:00 (6%), 3:00 (10%), 4:00 (6%), 5:00 (7%), 6:00 (7%), 7:00 (5%), 8:00 (7%), 9:00 (9%), 10:00 (6%), 11:00 (5%), and 12:00 (8%). The central region demonstrated smaller UM thickness ($p < .01$) and diameter ($p < .01$). The MFS (5- and 10-years) for central region (90%, 86%) was more favorable than any clock hour region ($p = .03$).

The UM anteroposterior regions included central (18%), post-equatorial (28%), equatorial (26%), peripheral (13%), and ciliary body (15%). Increasing UM thickness ($p < .01$) and diameter ($p < .01$) stepwise were noted with increasing anterior location. The MFS (5- and 10-years) for central region (90%, 86%) was more favorable than other anteroposterior regions ($p < .01$).

A combination of clock hour and anteroposterior regions (49 zones) revealed random distribution with no preferential location. Multivariate regression analysis adjusted for tumor thickness/diameter and proximity to foveola/optic disc showed no benefit of any location.

Conclusions: Across all fundus regions, UM occurred most often in the central region. Uveal melanoma in the central region was smaller in thickness and base and demonstrated more favorable MFS compared to other quadrants, clock hours, and anteroposterior zones. Analysis of the 49 zones showed no benefit of tumor location alone after adjusting for tumor size and proximity to foveola and optic disc.

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Uveal melanoma (UM) can arise in any portion of the uveal tissue, ranging from iris to ciliary body to choroid, and this malignancy can portend a poor prognosis.^{1,2} In 1982, McLean et al. studied 3,432 cases of UM managed with enucleation and submitted to the Armed Forces Institute of Pathology (AFIP), noting that those with ciliary body or choroidal involvement demonstrated far greater risk for metastasis than iris melanoma.³ They speculated that ciliary body and choroidal melanoma were of greater size and more malignant cytology than iris melanoma; hence, tumor location was an important prognostic factor.³ In 1995, McLean et al. revisited the impact of tumor location of ciliary body and choroidal melanoma in 664 globes submitted to the AFIP and noted ciliary body involvement was associated with greater melanoma-related death but commented that this was likely due to larger tumor size and more malignant cytopathology, than location.⁴ Others have correlated greater melanoma-related death with anatomic tumor location in the ciliary body.^{5,6}

There is limited literature regarding specific UM location within the eye or its impact on prognosis. Li et al. reviewed

tumor location of UM in 448 cases and noted a heavy concentration of UM in the macular region ($p < .01$) but found no evidence of clock hour or zonal preference.⁷ They did not correlate specific location with metastatic risk. They concluded that UM location is not uniformly distributed, concentrated in the macula, and with decreasing frequency towards the periphery, consistent with the influence of solar light on the development of this malignancy.⁷ Herein, we evaluate specific UM location in a larger cohort graded by quadrant, clock hour, anteroposterior region, and a combination of clock hour plus anteroposterior region to better understand the frequency of tumor location in specific zones and the impact of tumor location on metastasis-free survival (MFS).

METHODS

The medical records of 1001 patients diagnosed with UM between February 2004 and June 2020 from the Ocular Oncology Service at Wills Eye Hospital, Thomas Jefferson University, Philadelphia, Pennsylvania, USA were retrospectively reviewed and evaluated.

Approval by the Institutional Review Board was obtained and the study was conducted in accordance with the tenets of the Declaration of Helsinki and the Health Insurance Portability and Accountability Act. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) protocols were followed.⁸ Inclusion criteria included availability of specific tumor location and tumor cytogenetics. Informed consent was obtained from all patients.

All patients underwent a comprehensive clinical evaluation by an ocular oncology team (JAS, CLS, SEL) to establish the diagnosis, with examination of the anterior and posterior segment of the eye using slit lamp biomicroscopy, indirect ophthalmoscopy, and gonioscopy and transillumination as needed. Multimodal imaging was performed at initial and subsequent examinations and included fundus photography, fundus autofluorescence, ultrasonography (A- and B-scan), optical coherence tomography, and fluorescein angiography and indocyanine green angiography as needed. Data collection from medical records consisted of demographic features (age, race, sex, affected eye), clinical features (tumor epicenter location, tumor thickness and diameter, and tumor distance to the foveola and optic disc). All patients received treatment (plaque radiotherapy, stereotactic radiotherapy, enucleation, or resection).

The tumor epicenter was accurately determined by superimposing a transparent retinal diagram with an equidistant clock hour projection over the large fundus drawing of the

UM that was approved by ocular oncologists (JAS, CLS, SEL).^{9,10} (Figure 1) The tumor epicenter location was then categorized into 4 independent regions including quadrant, clock hour, anteroposterior, and a combination of clock hour and anteroposterior regions. (Figure 2) The tumor quadrant regions comprised 5 zones (central (macula and juxtapapillary), superior (10:30–1:30 for both eyes (OU)), nasal (1:30–4:30 for right eye (OD) and 7:30–10:30 for left eye (OS)), inferior (4:30–7:30 for OU), and temporal (7:30–10:30 for OD and 1:30–4:30 for OS)). (Figure 2A) The clock hour regions comprised 13 zones (central, 1:00, 2:00, 3:00, 4:00, 5:00, 6:00, 7:00, 8:00, 9:00, 10:00, 11:00, and 12:00). (Figure 2C) The anteroposterior regions comprised 5 zones [central, post-equatorial, equatorial, peripheral, and ciliary body (including those with iris extension)]. (Figure 2E) The combination of clock hour and anteroposterior regions comprised 49 zones. (Figure 2G) The percentage of cases in each zone was recorded (Figures 2A, C, E, G) and a frequency heat map for each region (up to maximum 30%) was constructed (Figures 2B, D, F, H) to illustrate the tumor location in color (blue, green, yellow, orange, red from least to greatest frequency).

Following treatment (plaque brachytherapy or enucleation or partial lamellar sclerouvectomy or stereotactic radiation), the patients were monitored 2–3 times yearly from an ocular and systemic standpoint. Systemic melanoma-related metastasis was documented by a medical oncologist or local medical

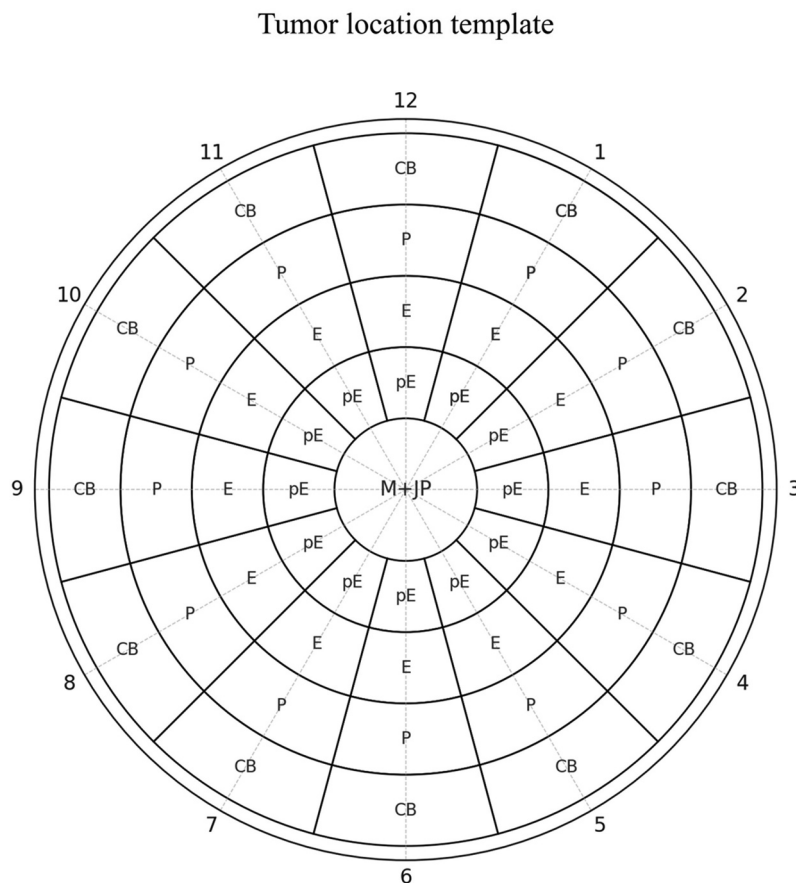


Figure 1. Tumor location template. Fundus diagram representing combination of tumor location in clock hour plus anteroposterior regions comprising 49 zones. (M-macula and JP-juxtapapillary [referred to as central], pE-post-equatorial, E-equatorial, P-peripheral, CB-ciliary body).

Tumor location by quadrant, clock hour, anteroposterior, and combination clock hour and anteroposterior regions

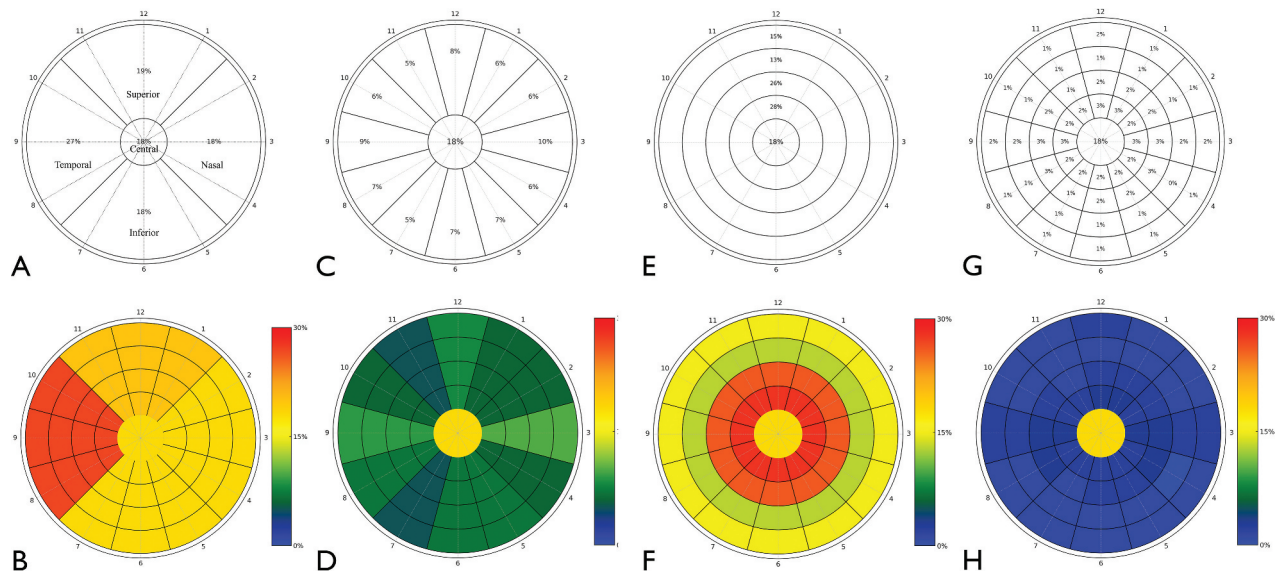
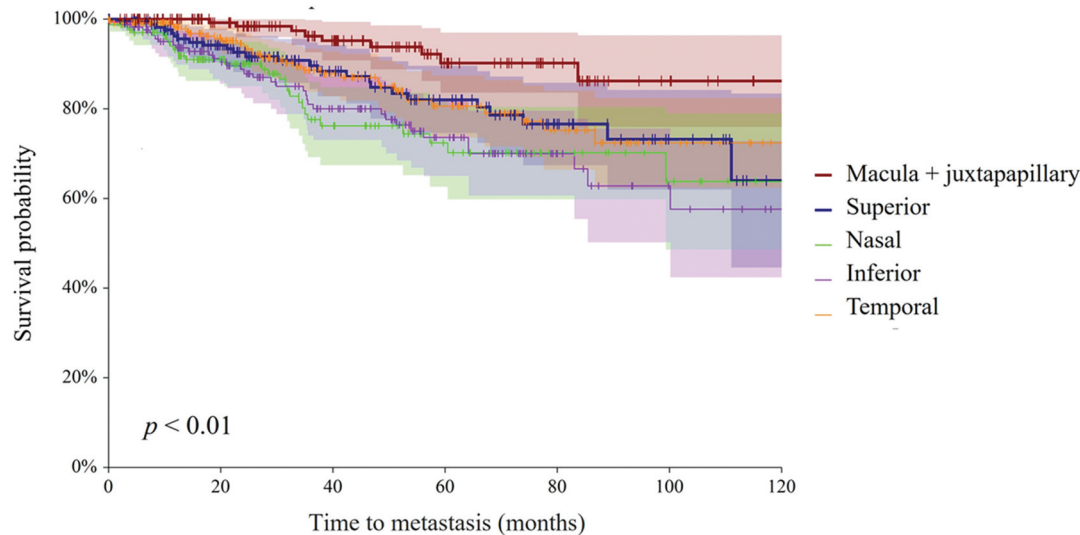


Figure 2. Tumor location by 4 different regions. (A) Quadrant regions (5 zones) showing labeled tumor frequency and (B) heat map demonstrating greatest frequency temporally. (C) Clock hour regions (13 zones) showing labeled tumor frequency and (D) heat map demonstrating greatest frequency in central zone. (E) Anteroposterior regions (5 zones) showing labeled tumor frequency and (F) heat map demonstrating greatest frequency in post-equatorial and equatorial zones. (G) Combination of clock hour plus anteroposterior regions (49 zones) showing labeled tumor frequency and (H) heat map demonstrating greatest frequency in central zone.

Kaplan-Meier survival estimates of uveal melanoma in 1001 cases by location of quadrant regions



Macula + juxtapapillary	180	120	82	46	24	9	3
Superior	195	123	75	54	29	14	3
Nasal	176	98	54	33	18	9	3
Inferior	184	110	76	47	23	12	6
Temporal	266	176	103	63	34	13	5

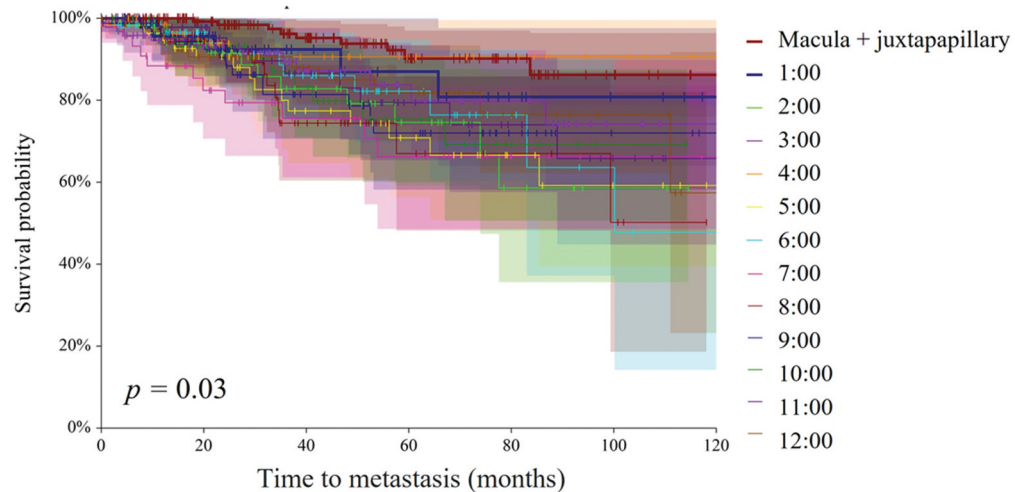
Figure 3. Kaplan-Meier survival estimates based on tumor location by quadrant regions. Central (macula + juxtapapillary) region demonstrated most favorable 10-year survival ($p < .01$).

doctor via imaging modalities including magnetic resonance imaging, computed tomography, or ultrasonography in conjunction with either histopathologic or cytopathologic confirmation. Information regarding metastasis and death was gathered based on medical history from the medical care team.

Statistical analysis

Statistical analysis was performed using NCSS version 21 software. The Shapiro-Wilk normality test was used to assess the distribution of variables. Continuous variables were presented

Kaplan-Meier survival estimates of uveal melanoma in 1001 cases by location of clock hour regions



Macula + juxtapapillary	180	120	82	46	24	9	3
1:00	58	34	18	14	8	4	1
2:00	61	39	26	15	6	2	0
3:00	100	67	34	21	15	6	3
4:00	61	38	23	16	11	7	4
5:00	67	43	27	18	10	6	3
6:00	68	39	30	15	8	4	2
7:00	49	28	19	14	5	2	1
8:00	72	42	20	9	5	3	0
9:00	86	51	30	21	10	3	1
10:00	62	37	24	14	5	1	0
11:00	54	40	26	16	10	5	1
12:00	83	49	31	24	11	5	1

Figure 4. Kaplan-Meier survival estimates based on tumor location by clock hour regions. Central (macula + juxtapapillary), 4:00, and 1:00 region demonstrated most favorable 10-year survival ($p = .03$).

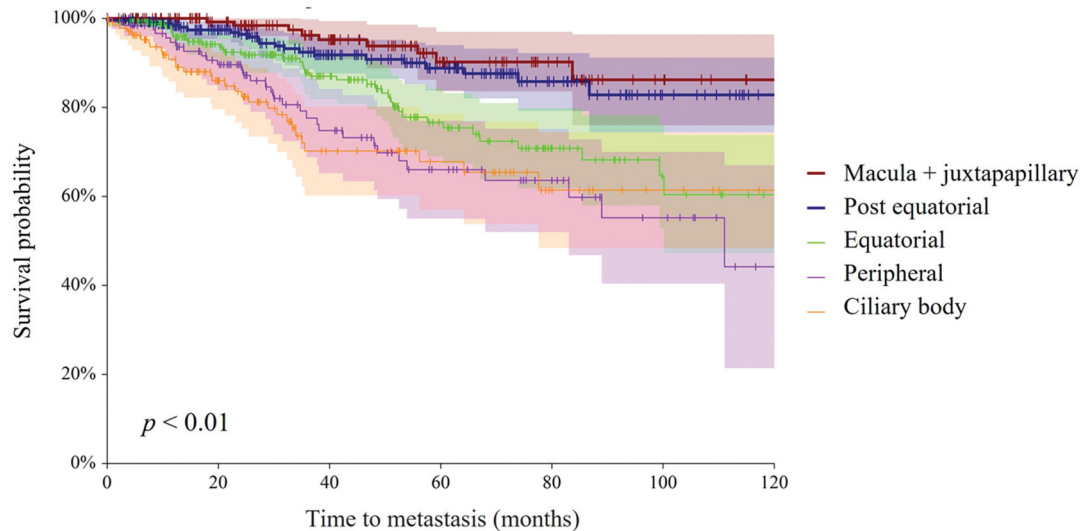
as median (25%-75% interquartile range) and mean $\pm$ standard deviation. Statistical comparisons were performed using appropriate tests: Kruskal-Wallis test for non-normally distributed continuous variables, Pearson chi-squared test for nominal-ordinal data, and post hoc tests where appropriate. The primary outcomes were two-fold, namely tumor location and impact of tumor location on melanoma-related metastasis using cumulative incidence of metastasis (CIM) per each quadrant (5 zones for each eye), radial clock hours (13 zones), anteroposterior (5 zones), and circumferential (clock hour plus anteroposterior regions) (49 zones) regions. Kaplan-Meier survival estimates were performed to assess MFS, followed by the log-rank test and Gray's test to evaluate the statistical significance of differences observed between the groups. Our threshold for statistical significance was set at $p < .05$, with adjusted p values reported where necessary to account for multiple comparisons. Circular heat maps are radial two-dimensional maps, wherein individual values are contained in a matrix represented by individual colors based on incidence of a specific outcome.^{11,12} In this study, the heat maps were plotted based on tumor location, thus giving a color visualization of the data frequency.

RESULTS

There were 1001 consecutive patients with UM who demonstrated specific tumor location and were sampled for cytogenetic testing at the Ocular Oncology Service, Wills Eye Hospital, Philadelphia, PA USA. At presentation this cohort demonstrated mean age of 58 years (median 59, range 10–94 years), white race ($n = 965$, 96%), male sex ($n = 516$, 52%), and affected right eye ($n = 529$, 53%). The UM features included tumor thickness (mean 6, median 5 mm), diameter (mean 12, median 12 mm) and with tumor distance to the foveola (mean 4, median 3 mm) and optic disc (mean 5, median 4 mm).

The tumor epicenter location by quadrant, clock hour, and anteroposterior regions per demographics is listed in Table 1 and per clinical features is listed in Table 2. Tumor location by quadrant region included 5 zones comprising central (18%), superior (19%), nasal (18%), inferior (18%), and temporal (27%). The central zone revealed patients more frequently younger with mean patient age ($p < .01$), male sex ($p = .02$), with smaller tumor thickness ($p < .01$) and diameter ($p < .01$), and shorter distance to the foveola ($p < .01$) and optic disc ($p < .01$). There was no difference in patient race or affected eye per quadrant zone.

Kaplan-Meier survival estimates of uveal melanoma in 1001 cases by location of anteroposterior regions



Macula + juxtapapillary	180	120	82	46	24	9	3
Post equatorial	284	185	117	78	38	16	4
Equatorial	259	159	101	60	32	16	9
Peripheral	129	86	51	32	21	10	2
Ciliary body	149	77	39	27	13	6	2

Figure 5. Kaplan-Meier survival estimates based on tumor location by anteroposterior regions. Central (macula + juxtapapillary) and post-equatorial regions demonstrated most favorable 10-year survival ($p < .01$).

Table 1. Tumor location of uveal melanoma by quadrant, clock hour, and anteroposterior regions. Demographics.

Tumor epicenter location	Age (in years) [Median (IQR 25% – 75%)]	Race (White/non-White)	Sex (male/female)	Affected eye (right/left)
By quadrant region				
Central* (n = 180, 18%)	57 (47–65)	173 (96%)/7 (4%)	112 (62%)/68 (38%)	87 (48%)/93 (52%)
Superior (n = 195, 19%)	60 (51–70)	187 (96%)/8 (4%)	93 (48%)/102 (52%)	108 (55%)/87 (45%)
Nasal (n = 176, 18%)	61 (53–70)	172 (98%)/4 (2%)	96 (55%)/80 (45%)	96 (55%)/80 (45%)
Inferior (n = 184, 18%)	60 (48–69)	180 (98%)/4 (2%)	88 (48%)/96 (52%)	98 (53%)/86 (47%)
Temporal (n = 266, 27%)	61 (51–70)	253 (95%)/13 (5%)	127 (48%)/139 (52%)	140 (53%)/126 (47%)
p-value	<0.01	0.48	0.02	0.66
By clock hour region				
Central* (n = 180, 18%)	57 (47–65)	173 (96%)/7 (4%)	112 (62%)/68 (38%)	87 (48%)/93 (52%)
1:00 (n = 58, 6%)	62 (52–71)	56 (97%)/2 (3%)	25 (43%)/33 (57%)	31 (53%)/27 (47%)
2:00 (n = 61, 6%)	64 (56–70)	59 (97%)/2 (3%)	27 (44%)/34 (56%)	31 (51%)/30 (49%)
3:00 (n = 100, 10%)	59 (46–70)	97 (97%)/3 (3%)	57 (57%)/43 (43%)	38 (38%)/62 (62%)
4:00 (n = 61, 6%)	63 (54–69)	55 (90%)/6 (10%)	30 (49%)/31 (51%)	27 (44%)/34 (56%)
5:00 (n = 67, 7%)	60 (48–65)	67 (100%)/0 (0%)	33 (49%)/34 (51%)	34 (51%)/33 (49%)
6:00 (n = 68, 7%)	60 (47–72)	66 (97%)/2 (3%)	34 (50%)/34 (50%)	37 (54%)/31 (46%)
7:00 (n = 49, 5%)	61 (51–69)	47 (96%)/2 (4%)	21 (43%)/28 (57%)	27 (55%)/22 (45%)
8:00 (n = 72, 7%)	61 (55–69)	71 (99%)/1 (1%)	33 (46%)/39 (54%)	42 (58%)/30 (42%)
9:00 (n = 86, 9%)	59 (47–68)	82 (95%)/4 (5%)	44 (51%)/42 (49%)	58 (67%)/28 (33%)
10:00 (n = 62, 6%)	61 (54–70)	61 (98%)/1 (2%)	32 (52%)/30 (48%)	40 (65%)/22 (35%)
11:00 (n = 54, 5%)	57 (45–65)	52 (96%)/2 (4%)	26 (48%)/28 (52%)	30 (56%)/24 (44%)
12:00 (n = 83, 8%)	60 (52–70)	79 (95%)/4 (5%)	42 (51%)/41 (49%)	47 (57%)/36 (43%)
p-value	<0.01	0.48	0.25	0.02
By anteroposterior region				
Central* (n = 180, 18%)	57 (47–65)	173 (96%)/7 (4%)	112 (62%)/68 (38%)	87 (48%)/93 (52%)
Post-equatorial (n = 284, 28%)	52 (50–69)	277 (98%)/7 (2%)	145 (51%)/139 (49%)	144 (51%)/140 (49%)
Equatorial (n = 259, 26%)	60 (52–69)	251 (97%)/8 (3%)	130 (50%)/129 (50%)	146 (56%)/113 (44%)
Peripheral (n = 129, 13%)	62 (52–71)	122 (95%)/7 (5%)	67 (52%)/62 (48%)	64 (50%)/65 (50%)
Ciliary body (n = 149, 15%)	60 (49–70)	142 (95%)/7 (5%)	62 (42%)/87 (58%)	88 (59%)/61 (41%)
p-value	<0.01	0.68	0.01	0.13

*Central = macula + juxtapapillary.

Table 2. Tumor location of uveal melanoma by quadrant, clock hour, and anteroposterior regions. Clinical features.

Tumor epicenter location	Tumor thickness (mm) [Median (IQR 25% – 75%)]	Tumor diameter (mm) [Median (IQR 25% – 75%)]	Distance to foveola (mm) [Median (IQR 25% – 75%)]	Distance to optic disc (mm) [Median (IQR 25% – 75%)]
By quadrant region				
Central*	3.8 ± 2.4	8.7 ± 3.4	0.7 ± 1.1	1.0 ± 1.4
(n = 180, 18%)	3.0 (2.3–4.5)	8.0 (6.0–10.0)	0.0 (0.0–1.0)	0.0 (0.0–2.0)
Superior	5.5 ± 3.1	12.4 ± 4.3	5.3 ± 4.0	5.2 ± 3.8
(n = 195, 19%)	4.8 (3.0–7.8)	12.0 (9.0–16.0)	4.0 (2.0–8.0)	4.0 (2.5–8.0)
Nasal	6.2 ± 3.0	13.0 ± 4.0	6.8 ± 4.1	4.0 ± 4.1
(n = 176, 18%)	6.0 (3.6–8.7)	13.0 (10.0–16.0)	6.0 (3.0–9.5)	3.0 (0.0–6.0)
Inferior	6.0 ± 3.2	13.2 ± 4.2	4.8 ± 4.2	5.1 ± 4.3
(n = 184, 18%)	5.4 (3.5–8.1)	14.0 (10.0–16.0)	4.0 (1.0–7.5)	4.0 (2.0–7.0)
Temporal	6.1 ± 3.4	12.8 ± 4.1	3.8 ± 3.8	6.2 ± 4.0
(n = 266, 27%)	5.0 (3.4–8.6)	13.0 (10.0–16.0)	3.0 (1.0–6.0)	5.5 (3.2–9.0)
p-value	<0.01	<0.01	<0.01	<0.01
By clock hour region				
Central*	3.8 ± 2.4	8.7 ± 3.4	0.7 ± 1.1	1.0 ± 1.4
(n = 180, 18%)	3.0 (2.3–4.5)	8.0 (6.0–10.0)	0.0 (0.0–1.0)	0.0 (0.0–2.0)
1:00	5.5 ± 3.3	12.1 ± 4.1	5.4 ± 3.9	5.0 ± 3.8
(n = 58, 6%)	4.1 (3.0–7.7)	11.5 (9.0–16.0)	4.0 (3.0–7.0)	4.0 (2.5–6.0)
2:00	6.8 ± 3.5	12.8 ± 4.6	4.8 ± 3.6	4.5 ± 3.9
(n = 61, 6%)	6.4 (3.6–9.9)	12.0 (9.0–17.0)	4.0 (3.0–6.5)	4.0 (1.0–7.0)
3:00	5.9 ± 3.0	12.7 ± 4.0	5.2 ± 4.3	5.7 ± 4.0
(n = 100, 10%)	5.5 (3.4–8.7)	12.0 (9.2–16.0)	4.0 (2.0–8.0)	5.5 (3.0–8.0)
4:00	5.7 ± 3.2	12.6 ± 3.6	4.9 ± 4.2	4.8 ± 4.4
(n = 61, 6%)	4.7 (3.2–7.6)	12.0 (10.0–16.0)	4.0 (1.0–8.0)	4.0 (0.3–8.0)
5:00	5.5 ± 2.7	12.5 ± 3.9	4.8 ± 4.5	5.2 ± 4.7
(n = 67, 7%)	5.0 (3.2–7.4)	12.0 (9.0–16.0)	3.0 (1.7–8.0)	4.0 (2.0–7.0)
6:00	6.7 ± 3.5	14.0 ± 4.3	4.6 ± 4.2	4.9 ± 4.0
(n = 68, 7%)	6.0 (3.7–9.6)	14.0 (11.0–11.7)	3.2 (1.7–7.0)	4.4 (2.0–7.0)
7:00	5.8 ± 3.3	13.1 ± 4.3	5.1 ± 4.0	5.2 ± 4.1
(n = 49, 5%)	5.5 (3.0–7.0)	13.0 (10.0–16.0)	5.0 (1.2–8.0)	4.0 (2.0–8.0)
8:00	6.3 ± 3.7	13.1 ± 3.9	5.3 ± 4.5	5.4 ± 4.7
(n = 72, 7%)	4.9 (3.5–8.6)	13.0 (10.0–16.0)	4.2 (1.0–9.0)	4.0 (1.1–8.7)
9:00	6.7 ± 3.4	13.6 ± 4.4	5.0 ± 4.5	5.9 ± 4.3
(n = 86, 9%)	6.4 (3.5–10.0)	14.0 (10.0–17.0)	3.0 (2.0–8.0)	5.5 (3.0–8.3)
10:00	5.3 ± 2.5	12.0 ± 3.6	4.8 ± 3.9	5.3 ± 3.7
(n = 62, 6%)	4.9 (3.2–6.4)	12.0 (9.0–15.0)	4.0 (1.5–8.0)	4.2 (3.0–8.0)
11:00	5.6 ± 2.5	13.2 ± 4.1	5.1 ± 4.4	5.3 ± 4.0
(n = 54, 5%)	5.4 (3.6–6.9)	14.0 (9.7–16.0)	4.0 (2.0–8.0)	4.0 (2.0–8.0)
12:00	5.5 ± 3.3	12.2 ± 4.5	5.4 ± 3.9	5.3 ± 3.7
(n = 83, 8%)	5.0 (2.7–8.2)	12.0 (8.0–16.0)	4.0 (2.3–8.0)	4.2 (2.5–8.0)
p-value	<0.01	<0.01	<0.01	<0.01
By anteroposterior region				
Central*	3.8 ± 2.4	8.7 ± 3.4	0.7 ± 1.1	1.0 ± 1.4
(n = 180, 18%)	3.0 (2.3–4.5)	8.0 (6.0–10.0)	0.0 (0.0–1.0)	0.0 (0.0–2.0)
Post-equatorial	4.6 ± 2.4	10.9 ± 3.6	2.1 ± 1.9	2.7 ± 2.2
(n = 284, 28%)	4.0 (3.0–5.6)	11.0 (8.0–14.0)	2.0 (0.5–3.0)	3.0 (0.5–4.0)
Equatorial	5.9 ± 3.0	13.3 ± 3.9	5.0 ± 2.9	5.0 ± 3.2
(n = 259, 26%)	5.2 (3.6–7.8)	14.0 (10.0–16.0)	5.0 (3.0–7.0)	5.0 (3.0–7.0)
Peripheral	7.1 ± 3.5	14.7 ± 3.7	7.6 ± 4.0	7.5 ± 3.9
(n = 129, 13%)	6.4 (4.0–10.1)	16.0 (12.0–18.0)	8.0 (4.0–11.0)	7.0 (5.0–10.0)
Ciliary body	7.6 ± 3.6	14.0 ± 4.5	10.0 ± 4.6	10.2 ± 4.6
(n = 149, 15%)	7.9 (4.7–10.1)	15.0 (11.0–18.0)	11.5 (6.0–13.0)	11.0 (7.0–14.0)
p-value	<0.01	<0.01	<0.01	<0.01

*Central = macula + juxtapapillary.

Tumor location by clock hour regions included 13 zones comprising central (18%), 1:00 (6%), 2:00 (6%), 3:00 (10%), 4:00 (6%), 5:00 (7%), 6:00 (7%), 7:00 (5%), 8:00 (7%), 9:00 (9%), 10:00 (6%), 11:00 (5%), and 12:00 (8%). (Tables 1 and 2) The central region revealed younger mean patient age ($p < .01$), with smaller tumor thickness ($p < .01$) and diameter ($p < .01$), and shorter distance to the foveola ($p < .01$) and optic disc ($p < .01$). There was no difference in race or sex per clock hour zone.

Tumor location by anteroposterior regions included 5 zones comprising central (18%), post-equatorial (28%), equatorial (26%), peripheral (13%), and ciliary body (15%). (Tables 1 and 2) The central region revealed younger mean patient age ($p < .01$) and male sex ($p = .01$), and increasing anterior location demonstrated stepwise increasing tumor thickness (p

$< .01$) and diameter ($p < .01$), and longer distance to the foveola ($p < .01$) and optic disc ($p < .01$). There was no difference in race or affected eye per anteroposterior zone.

Outcomes of MFS (5- and 10-years) are listed in Table 3. The MFS by quadrant region (Figure 3) revealed central zone (90%, 86%) was more favorable than other quadrants ($p < .01$). The MFS by clock hour region (Figure 4) revealed central zone (90%, 86%) was more favorable than other clock hour zones ($p = .03$). The MFS by anteroposterior region (Figure 5) revealed central zone (90%, 86%) was more favorable than other anteroposterior regions ($p < .01$).

A comparative analysis of UM tumor location in the 49 zones representing a combination of clock hour and anteroposterior regions is listed in Table 4. There was random

Table 3. Tumor location of uveal melanoma and impact on metastasis-free survival in 1001 cases: Kaplan-Meier survival estimates.

Tumor epicenter location	Cumulative incidence of metastasis [n (%)]	Kaplan-Meier survival estimate (CI 95%)				p-value
		1-year	3-years	5-years	10-years	
By quadrant region						
Central* (n = 180, 18%)	10 (6%)	100%	96% (92–100)	90% (83–97)	86% (76–96)	<0.01
Superior (n = 195, 19%)	25 (13%)	96% (93–99)	90% (84–95)	82% (74–90)	64% (45–83)	
Nasal (n = 176, 18%)	29 (16%)	93% (89–97)	78% (70–86)	73% (63–82)	64% (49–79)	
Inferior (n = 184, 18%)	35 (19%)	94% (90–97)	81% (74–88)	74% (65–82)	58% (42–73)	
Temporal (n = 266, 27%)	32 (12%)	98% (97–100)	89% (84–93)	80% (74–87)	72% (62–82)	
p-value	<0.01					
By clock hour region						
Central* (n = 180, 18%)	10 (6%)	100%	96% (92–100)	90% (83–97)	86% (76–96)	0.03
1:00 (n = 58, 6%)	5 (9%)	95% (89–100)	92% (84–100)	87% (74–100)	81% (64–98)	
2:00 (n = 61, 6%)	11 (18%)	96% (91–100)	83% (71–95)	74% (59–90)	NA	
3:00 (n = 100, 10%)	12 (12%)	94% (90–99)	89% (82–98)	84% (74–94)	74% (58–90)	
4:00 (n = 61, 6%)	4 (6%)	98% (95–100)	91% (82–100)	91% (82–100)	91% (82–100)	
5:00 (n = 67, 7%)	14 (21%)	94% (88–100)	80% (68–92)	71% (56–85)	59% (40–79)	
6:00 (n = 68, 7%)	10 (15%)	96% (92–100)	86% (75–97)	82% (70–95)	48% (14–81)	
7:00 (n = 49, 5%)	11 (22%)	88% (79–98)	75% (61–90)	66% (49–84)	66% (49–84)	
8:00 (n = 72, 7%)	12 (17%)	96% (91–100)	75% (61–89)	68% (50–86)	NA	
9:00 (n = 86, 9%)	13 (15%)	96% (91–100)	81% (70–92)	72% (58–86)	72% (58–86)	
10:00 (n = 62, 6%)	9 (14%)	98% (93–100)	87% (76–98)	75% (60–91)	NA	
11:00 (n = 54, 5%)	9 (17%)	98% (93–100)	87% (75–98)	79% (65–93)	66% (45–87)	
12:00 (n = 83, 8%)	11 (13%)	96% (91–100)	91% (83–98)	82% (70–93)	57% (23–92)	
p-value	0.02					
By anteroposterior region						
Central* (n = 180, 18%)	10 (6%)	100%	96% (92–100)	90% (83–97)	86% (76–96)	<0.01
Post-equatorial (n = 284, 28%)	21 (7%)	98% (97–100)	92% (88–96)	89% (84–94)	83% (74–91)	
Equatorial (n = 259, 26%)	39 (15%)	96% (93–98)	88% (83–93)	76% (69–84)	60% (47–74)	
Peripheral (n = 129, 13%)	21 (16%)	95% (90–99)	78w% (69–86)	66% (55–77)	44% (21–67)	
Ciliary body (n = 149, 15%)	30 (20%)	91% (86–96)	70% (60–80)	68% (57–78)	61% (48–74)	
p-value	<0.01					

*Central = macula + juxtapapillary.

Table 4. Comparative analysis of cumulative incidence of metastasis in 49 fundus zones based on combination of clock hour and anteroposterior regions.

Location variables	Central*	Post-equator	Equator	Periphery	Ciliary body	p-value
Central*	n _{all} = 180 (18%) n _{CIM} = 10 (6%)	NA	NA	NA	NA	NA
1:00	NA	n _{all} = 26 (3%) n _{CIM} = 1 (4%)	n _{all} = 16 (2%) n _{CIM} = 2 (12%)	n _{all} = 7 (1%) n _{CIM} = 0 (0%)	n _{all} = 9 (1%) n _{CIM} = 2 (22%)	0.31
2:00	NA	n _{all} = 21 (2%) n _{CIM} = 2 (10%)	n _{all} = 18 (2%) n _{CIM} = 3 (17%)	n _{all} = 9 (1%) n _{CIM} = 1 (11%)	n _{all} = 13 (1%) n _{CIM} = 5 (38%)	0.20
3:00	NA	n _{all} = 33 (3%) n _{CIM} = 2 (6%)	n _{all} = 34 (3%) n _{CIM} = 3 (9%)	n _{all} = 15 (1%) n _{CIM} = 3 (20%)	n _{all} = 18 (2%) n _{CIM} = 4 (22%)	0.22
4:00	NA	n _{all} = 19 (2%) n _{CIM} = 0 (0%)	n _{all} = 32 (3%) n _{CIM} = 1 (3%)	n _{all} = 2 (1%) n _{CIM} = 0 (0%)	n _{all} = 8 (1%) n _{CIM} = 3 (38%)	<0.01
5:00	NA	n _{all} = 23 (2%) n _{CIM} = 0 (0%)	n _{all} = 18(2%) n _{CIM} = 6 (33%)	n _{all} = 13 (1%) n _{CIM} = 4 (31%)	n _{all} = 13 (1%) n _{CIM} = 4 (31%)	0.03
6:00	NA	n _{all} = 22 (2%) n _{CIM} = 2 (9%)	n _{all} = 19 (2%) n _{CIM} = 5 (26%)	n _{all} = 13 (1%) n _{CIM} = 2 (15%)	n _{all} = 14 (1%) n _{CIM} = 1 (7%)	0.43
7:00	NA	n _{all} = 17 (2%) n _{CIM} = 3 (18%)	n _{all} = 12 (1%) n _{CIM} = 2 (17%)	n _{all} = 9 (1%) n _{CIM} = 3 (33%)	n _{all} = 11 (1%) n _{CIM} = 3 (27%)	0.78
8:00	NA	n _{all} = 21 (2%) n _{CIM} = 2 (10%)	n _{all} = 27 (3%) n _{CIM} = 5 (19%)	n _{all} = 12 (1%) n _{CIM} = 3 (25%)	n _{all} = 12 (1%) n _{CIM} = 2 (17%)	0.69
9:00	NA	n _{all} = 26 (3%) n _{CIM} = 1 (4%)	n _{all} = 27 (3%) n _{CIM} = 7 (26%)	n _{all} = 16 (2%) n _{CIM} = 3 (19%)	n _{all} = 17 (2%) n _{CIM} = 2 (12%)	0.16
10:00	NA	n _{all} = 22 (2%) n _{CIM} = 1 (5%)	n _{all} = 21 (2%) n _{CIM} = 4 (19%)	n _{all} = 12 (1%) n _{CIM} = 2 (25%)	n _{all} = 7 (1%) n _{CIM} = 1 (14%)	0.39
11:00	NA	n _{all} = 22 (2%) n _{CIM} = 2 (9%)	n _{all} = 13 (1%) n _{CIM} = 1 (8%)	n _{all} = 7 (1%) n _{CIM} = 5 (71%)	n _{all} = 12 (1%) n _{CIM} = 1 (8%)	<0.01
12:00	NA	n _{all} = 32 (3%) n _{CIM} = 5 (16%)	n _{all} = 22 (2%) n _{CIM} = 0 (0%)	n _{all} = 14 (1%) n _{CIM} = 4 (29%)	n _{all} = 15 (2%) n _{CIM} = 2 (13%)	0.14
p-value	NA	0.44	0.05	0.30	0.47	

*Central = macular + juxtapapillary.

n_{all} = Number of patients out of 1001 cases.n_{CIM} = Cumulative incidence of metastasis for n_{all}.

Table 5. Univariate regression analysis for metastasis-free survival per tumor location.

Location variables	HR (95% CI)	p-value
By quadrant regions		
Superior/Central*	2.43	.02
Nasal/Central	3.71	<.01
Inferior/Central	3.61	<.01
Temporal/Central	2.31	.02
By clock hour regions		
1:00/Central	1.85	.26
2:00/Central	3.51	<.01
3:00/Central	2.30	.05
4:00/Central	1.22	.74
5:00/Central	3.72	<.01
6:00/Central	2.88	.02
7:00/Central	4.47	<.01
8:00/Central	3.97	<.01
9:00/Central	3.13	<.01
10:00/Central	3.12	.01
11:00/Central	2.70	.03
12:00/Central	2.60	.03
By anteroposterior regions		
Post-equatorial/Central	1.36	.42
Equatorial/Post-equatorial	2.92	<.01
Peripheral/Equatorial	4.57	<.01
Ciliary body/Peripheral	4.93	<.01

*Central = macula + juxtapapillary; HR = hazard Ratio, CI = Confidence Interval.

Table 6. Multivariate regression analysis for metastasis-free survival per tumor location, adjusted for tumor thickness, tumor diameter, distance to foveola, and distance to optic disc.

Location variables	HR (95% CI)	p-value
By quadrant regions		
Superior/Central*	0.98	.95
Nasal/Central	1.35	.20
Inferior/Central	1.20	.27
Temporal/Central	0.66	.06
By clock hour regions		
1:00/Central	0.25	.99
2:00/Central	0.27	.99
3:00/Central	0.16	.99
4:00/Central	0.09	.99
5:00/Central	0.38	.99
6:00/Central	0.19	.99
7:00/Central	0.38	.99
8:00/Central	0.38	.99
9:00/Central	0.18	.99
10:00/Central	0.39	.99
11:00/Central	0.25	.99
12:00/Central	0.22	.99
By anteroposterior regions		
Post-equatorial/Central	1.22	.50
Equatorial/Post-equatorial	1.99	.22
Peripheral/Equatorial	1.95	.15
Ciliary body/Peripheral	1.52	.17

*Central = macula + juxtapapillary; HR = hazard Ratio, CI = Confidence Interval.

distribution, and each zone represented 1% – 3% of the entire cohort. The MFS for the 49 zones revealed that only 4:00 ciliary body and 11:00 peripheral zones demonstrated greater frequency of metastasis ($p < .01$).

The univariate regression analysis for hazard ratio (HR) for MFS is listed in Table 5. The univariate analysis by quadrant region revealed that the superior quadrant (HR: 2.43), nasal quadrant (HR: 3.71), inferior quadrant (HR: 3.61), and temporal quadrant (HR: 2.31) were all associated with significantly higher risks of metastasis compared to the central region ($p = .02$, $p < .01$, $p < .01$, and $p = .02$, respectively). The univariate analysis by

clock hour region showed that the 1 o'clock region (HR: 1.85) and the 4 o'clock region (HR: 1.22) did not show a statistically significant increase in risk compared to the central region ($p = .26$ and $p = .74$ respectively). In contrast, the 2 o'clock region (HR: 3.51, $p < .01$), the 3 o'clock region (HR: 2.30, $p = .05$), the 5 o'clock region (HR: 3.72, $p < .01$), the 6 o'clock region (HR: 2.88, $p = .02$), the 7 o'clock region (HR: 4.47, $p < .01$), the 8 o'clock region (HR: 3.97, $p < .01$), the 9 o'clock region (HR: 3.13, $p < .01$), the 10 o'clock region (HR: 3.12, $p = .01$), the 11 o'clock region (HR: 2.70, $p = .03$), and the 12 o'clock region (HR: 2.60, $p = .03$) were associated with significantly higher risks of metastasis compared

Table 7. Multivariate regression analysis for metastasis-free survival per tumor location, adjusted for tumor thickness and tumor diameter.

Location variables	HR (CI 95%)	p-value
By quadrant regions		
Superior/Central*	0.98	.95
Nasal/Central	1.22	.24
Inferior/Central	1.26	.15
Temporal/Central	0.78	.14
By clock-hour regions		
1:00/Central	0.83	.66
2:00/Central	1.13	.66
3:00/Central	0.80	.49
4:00/Central	0.48	.12
5:00/Central	1.57	.08
6:00/Central	0.76	.37
7:00/Central	1.71	.06
8:00/Central	1.33	.31
9:00/Central	0.66	.14
10:00/Central	1.73	.08
11:00/Central	1.06	.85
12:00/Central	0.97	.94
By anteroposterior regions		
Post-equatorial/Central	1.20	.50
Equatorial/Post-equatorial	1.40	.23
Peripheral/Equatorial	1.38	.18
Ciliary body/Peripheral	1.30	.12

*Central = macula + juxtapapillary; HR = hazard Ratio, CI = Confidence Interval.

to the central region. The univariate analysis by anteroposterior region revealed that the post-equatorial region (HR: 1.36) had no significant increase in risk compared to the central region ($p = .42$). However, the equator region (HR: 2.92), the peripheral region (HR: 4.57), and the ciliary body region (HR: 4.93) were associated with significantly higher risks of metastasis compared to the central region ($p = .02$, $p < .01$, and $p < .01$, respectively).

A multivariate regression analysis for HR for MFS is listed in Table 6. After adjusting for tumor thickness, diameter, distance to foveola, and distance to the optic disc, there was no location predictive of worse MFS. A multivariate regression analysis for HR for MFS is listed in Table 7 and after adjusting for tumor thickness and diameter there was no location predictive of worse MFS.

DISCUSSION

There are only a few reports on the topic of UM location as it impacts metastatic disease.³⁻⁷ Years ago, with collection of enucleation specimens at the AFIP, McLean et al. published two reports demonstrating that posterior UM (choroidal and ciliary body location) showed greater metastatic risk than anterior UM (iris location) and they speculated this was related to both greater tumor size and more malignant cytopathology with posterior malignancies.² Later they re-evaluated a larger cohort and noted that ciliary body melanoma was particularly at high-risk for metastatic disease, most likely related to large tumor size and malignant cytopathology rather than location.⁴ In 2000, Li et al. studied metastatic melanoma death rates by anatomic site in 1,848 UM after proton beam radiotherapy and found that ciliochoroidal melanoma had higher rates of metastatic death than melanomas involving the choroid only, and this increased with the proportion of tumor in the ciliary body

($p = .006$).⁵ In fact, > 75% ciliary body involvement increased the relative risk ratio for melanoma-related death by 2.30.⁵

Another study by Li et al., in 2000, explored more specific tumor location of UM in 448 cases involving the choroid or ciliary

body, and occasionally with extension into the iris.⁷ They classified tumor location according to 12 clock hour zones from 1:00 to 12:00 plus a foveal zone and then additionally anteroposteriorly in 6 zones including foveal, parafoveal, posterior, peripheral, anterior, and ciliary body zones. They found that UM was most often located in the foveal, parafoveal and posterior zones, and tumor location frequency decreased stepwise with anterior location, suggesting, as they proposed, that a decrease in solar illumination was associated with decreased tumor development.⁷ They also noted that the temporal parafoveal region demonstrated the most common tumor location, a region that they indicated had the highest solar exposure.⁷

In our report, we noted the highest prevalence of tumor epicenter location by quadrant (5 zones), clock hour (13 zones), and combination of clock hour plus anteroposterior regions (49 zones) was in the macula, similar to the report by Li et al.⁷ Comparable to their study, we also observed an increase in tumor prevalence in the temporal region, when compared to the other quadrants, further suggesting a heightened role for solar exposure in the development of UM (Figures 2A, 2B). However, in the anteroposterior region (5 zones), tumor location was most frequent in the post-equatorial and equatorial regions. (Figures 2E, 2F) In addition, when exploring the 49 zones divided by clock hour and anteroposterior regions, there was random distribution per zone, with no prevalence of any site, and all extramacular zones were similar. (Figures 2G, 2H)

We further explored the impact of tumor location on MFS, as there have been publications implying that ciliary body location was a higher risk for metastatic disease.^{3–5} Likewise, in the anteroposterior region analysis we found peripheral and ciliary body locations with poorest 5-year Kaplan-Meier MFS of 66% and 68%, respectively compared to central (91%), post-equatorial (89%) and equatorial (76%) ($p < .01$). However, when employing a multivariate regression analysis controlling for tumor size we found that tumor location was not an independent factor for MFS, but it was more likely related to tumor size. This could be pertaining to early detection of the central and post-equatorial regions as the patients tend to present early secondary to visual symptoms and tumors in these regions are relatively easy to detect on fundoscopic examination.

There are limitations to this analysis as it was retrospective and has inherent deficiencies in backdated data collection. However, every case was collected from meticulous chart documentation over several decades in a standardized fashion and tumor localization was graded by medical personnel using a systematized template created by the ocular oncology physicians to add precise accuracy for data collection. Fortunately, data collectors had both fundus drawing and wide-angle fundus imaging to use for identification of tumor location. However, location reflected only tumor epicenter and some tumors involved multiple zones. Further, data on melanoma-related metastasis was collected by medical personnel (medical oncologists, medical doctors) and patients to ensure accuracy in metastatic events.

In conclusion, in this analysis, we explored precise uveal melanoma location within the eye, studying quadrant (5 zones), clock hour (13 zones), and combination clock hour/anteroposterior (49 zones) regions and noted that central location demonstrated greatest frequency whereas in anteroposterior region (5 zones) post-equatorial and equatorial zones were most common tumor sites. Regarding all 4 regional assessments, the central region demonstrated the smallest tumors and MFS was most favorable; however, after adjusting for tumor thickness and diameter, there was no difference per location for MFS.

ABBREVIATIONS

AJCC	American Joint Committee on Cancer;
CIM	cumulative incidence of metastasis;
MFS	metastasis-free survival;
STROBE	Strengthening the Reporting of Observational Studies in Epidemiology;
TCGA	The Cancer Genome Atlas;
UM	uveal melanoma,
AFIP	Armed Forces Institute of Pathology.

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No potential conflict of interest was reported by the author(s).

DATA AVAILABILITY STATEMENT

Carol L. Shields, M.D., has had full access to all the data in the study and takes responsibility for the integrity of the data. The data sets generated and/or analyzed in the current study are not publicly available.

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